EXPERIMENT NO. 3

TITLE: - Measure the electrical bandwidth and optical bandwidth .Also compare it.

AIM: - To measure the bandwidth of optical fiber.

APPARATUS:-

1. Fiber optic analog communication kit (ST2502 trainer kit)
2. Fiber optic link
3. Oscilloscope

THEORY:-

In an analog communication link the transmitter contains either a LED or LASER Diode as optical source. Basically a fiber optic link contains three main elements, a transmitter, an optical fiber and a receiver. The transmitter module takes the input signal in electrical form and then transforms it into optical (light) energy containing the same information.

In order to transmit information in an optical fiber communication system, it is necessary to modulate a property of the light with the information signal. In this system the information signal is used to control the Intensity of the source. At the far end, the variation in the amplitude of the received signal is used to recover the original information signal.

The modulation bandwidth in optical communications may be defined in either electrical or optical terms. The modulation bandwidth is defined as the point where the electrical signal power has dropped to half its constant value resulting from the modulated portion of the optical signal.

Electrical bandwidth: The ratio of the electrical output power to the electrical input power in dB.

Electrical 3 dB bandwidth =10 log10 (output current/input current)

The frequency at which output power is reduced by 3 dB with respect to input power is called a 3dB electrical frequency.

Optical bandwidth: The ratio of the optical output power to the optical input power in dB.

Optical bandwidth = 10log10 (output current/input current)

Optical source exhibits a linear relationship between power and current. The frequency at which output power is reduced by half with respect to input power is called optical frequency.

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DIAGRAM:-

PROCEDURE:-

1. Connect the power supply cord to the main power plug & to trainer ST2502.
2. Ensure that all switched faults are “off”.
3. Connect the fiber optics cable between emitter output and detectors input.
4. Set input 2V or 3V and apply to the input of emitter.
5. Connect the detectors output to Oscilloscope.
6. On the board, switch emitter 1’s driver to analog mode.
7. Switch on the power supply of trainer and Oscilloscope.
8. Observe the output on Oscilloscope.
9. Vary the frequency in the range of 20HZ to 150KHZ and note the corresponding voltage.
10. Calculate voltage gain in dB in each case.
11. Plot the frequency response and find the 3 dB optical and electrical bandwidth

OBSERVATION TABLE:-

Vin = ----------

Sr.No.	Frequency (KHZ)	Output Voltage (V_0)	Gain $= 20 \log_{10}(V_0/V_{in})$

RESULT:-

3 dB electrical bandwidth (From Graph) =

Optical bandwidth (From Graph) =

CONCLUSION: